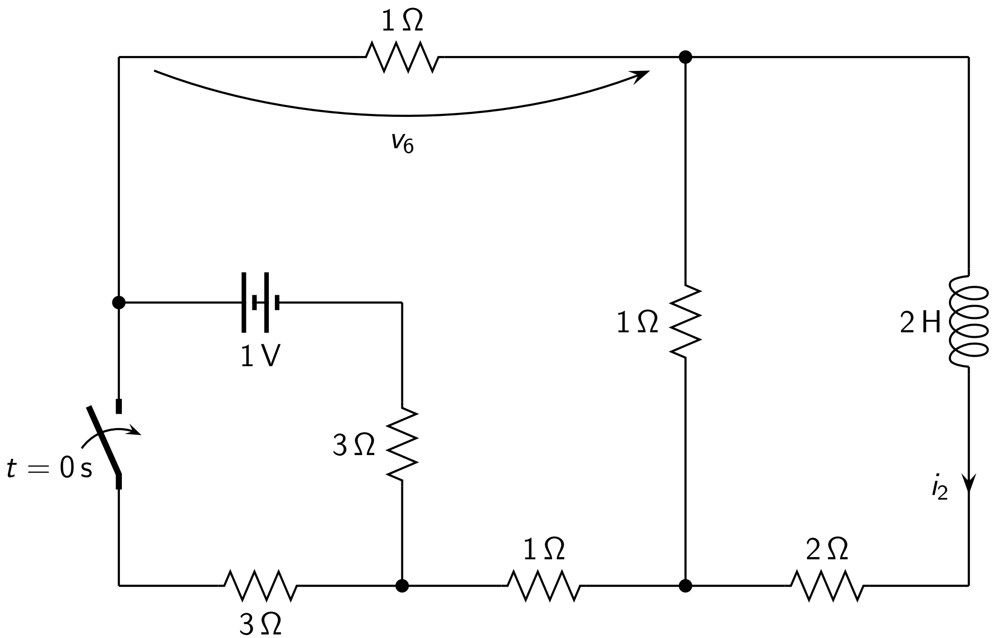
Problem: find the expression of $i_2(t)$, $v_6(t)$ for $t > 0$.



Solution

- Time constant: $\tau = \frac{18}{25}\,\text{s}$;
- Initial conditions before switching: $i_2(0^-) = \frac{1}{17}\,\text{A}$;
- Initial conditions after switching: $i_2(0^+) = \frac{1}{17}\,\text{A}$; $v_6(0^+) = -0.124\,\text{V}$;
- Steady-state solution: $i_2(\infty) = \frac{1}{25}\,\text{A}$; $v_6(\infty) = \frac{-3}{25}\,\text{V}$;
- Solution for $t > 0$:

$$i_2(t) = \left(18.824 \cdot 10^{-3}\, e^{-t/\tau} + \frac{1}{25}\right)\,\text{A}$$

$$v_6(t) = \left(-4.183 \cdot 10^{-3}\, e^{-t/\tau} + \frac{-3}{25}\right)\,\text{V}$$